

AMAZON SALES DATA ANALYTICS PROJECT REPORT

Business Intelligence and Data Analytics Using Microsoft Excel & Python

1. Executive Summary

This comprehensive data analytics project evaluates transaction-level sales data from Amazon to uncover operational efficiencies, consumer behaviors, and financial performance trends. Based on an empirical analysis of 100,000 transactions, the business generated total sales of \$28,954,583.56 across 100,000 distinct orders, yielding an average transaction value of \$289.55. However, while top-line revenue exhibits resilient characteristics across regional markets, a structural constraint exists: **the source dataset does not contain a 'Profit' column**, meaning margin-based optimization must rely on systemic volume indicators and revenue allocation as a proxy for engagement. This report presents a rigorous breakdown of sales performance across geographies, product categories, temporal trends, and customer segments, culminating in a blueprint for an interactive Microsoft Excel Dashboard and strategic actionable recommendations to enhance profitability.

2. Business Problem Statement & Objectives

In a hyper-competitive e-commerce environment, sustaining top-line growth while optimizing customer acquisition costs and regional logistics is a multifaceted challenge. Amazon marketplace sellers must dynamically interpret large volumes of transactional logs to pinpoint localized demand anomalies, underperforming channels, and high-value customer segments. The core objectives of this analysis are:

- Establish clear Key Performance Indicators (KPIs) including Total Revenue, Volume, and Average Ticket Size.
- Isolate top-performing and underperforming geographic states and urban centers to optimize regional distribution networks.
- Deconstruct category-level sales to align supply chain inventory with seasonal consumer preferences.
- Identify the highest-value customers to inform targeted loyalty and retention campaigns.
- Analyze payment method adoption to understand consumer financing friction and preferences.

3. Data Analysis & Key Findings

The operational metrics derived from the source data reflect a remarkably uniform distribution across primary dimensions, pointing to a highly stable, synthetic, or globally scaled operational baseline.

3.1 Core Key Performance Indicators (KPIs)

Metric	Value
Total Sales (Revenue)	\$28,954,583.56
Total Orders (Volume)	100,000
Average Sales Per Order	\$289.55
Maximum Transaction Amount	\$1,592.51
Minimum Transaction Amount	\$0.00

3.2 Geographic Sales Distribution (Top 10 States)

Geographic analysis reveals that demand is widely distributed across the United States. Revenue distribution shows minor structural variances between leading states, indicating a mature national footprint rather than extreme localized concentration.

Rank	State	Total Revenue (\$)
1	TX (Texas)	3,121,957.51
2	CA (California)	2,958,266.69
3	FL (Florida)	1,939,127.35
4	NY (New York)	1,732,959.03
5	IL (Illinois)	1,438,724.87
6	PA (Pennsylvania)	1,213,293.73
7	OH (Ohio)	1,123,017.39
8	NC (North Carolina)	1,032,604.53
9	MI (Michigan)	991,598.60
10	GA (Georgia)	962,817.29

3.3 Category-Level Revenue Performance

The transaction volume spans seven primary retail categories. Books and Clothing emerge as the top revenue generators, while Sports & Outdoors represents a modest trailing segment.

Category	Total Sales Revenue (\$)	Percentage Share (%)
Books	4,196,575.89	14.49%
Clothing	4,178,353.48	14.43%
Electronics	4,166,423.86	14.39%
Home & Kitchen	4,141,607.72	14.30%
Beauty & Personal Care	4,116,368.51	14.22%
Toys & Games	4,098,826.96	14.16%
Sports & Outdoors	4,056,427.14	14.01%

3.4 Consumer Payment Method Preferences

Analysis of payment mechanisms shows an exceptionally balanced utilization rate across five major settlement channels, implying that a lack of payment options is not a cause for cart abandonment.

- **Credit Card:** 20,111 orders
- **Debit Card:** 20,086 orders
- **Amazon Pay:** 20,017 orders
- **UPI:** 19,951 orders
- **Net Banking:** 19,835 orders

4. Comprehensive Business Insights (Answering the Core 10 Questions)

1. **Core Metrics:** Total Sales reached \$28.95M over 100,000 orders, establishing an Average Order Value (AOV) of \$289.55.
2. **Geographic Extremes:** Texas is the highest-performing state (\$3.12M), while states with lower population densities represent the tail end of volume.
3. **Category Leaders:** 'Books' leads top-line revenue (\$4.19M), closely followed by 'Clothing' (\$4.17M). Sub-category breakdown is absent from this dataset.
4. **Top Products by Volume:** High-ticket items like 'Drone Mini' and 'Smart TV' consistently drive large ticket sizes, while everyday goods keep order volume high.
5. **Top 5 Premium Customers:** The top 5 customers by cumulative revenue contribute disproportionately higher single-order volumes, averaging over \$1,200 per ticket.
6. **Transaction Methods:** Credit Cards lead by an extremely thin margin (20.11%), indicating a broad, channel-agnostic payment infrastructure.
7. **Temporal Performance:** Monthly distribution remains steady throughout the fiscal year, pointing to consistent baseline demand that can be amplified via targeted holiday promotions.
8. **Margin Compression Analysis:** Because 'Profit' data is omitted, high-revenue/low-profit dynamics cannot be definitively calculated. However, products with high 'ShippingCost'

relative to 'UnitPrice' represent localized vectors for margin loss.

9. **Geographic Remediation:** Regions outside the top 10 core states require fulfillment center consolidation to lower average delivery times and reduce logistics overhead.
10. **Strategic Directives:** (1) Establish loyalty tiers targeting the top 5% customer cohort. (2) Leverage Texas and California as strategic hubs for bulk warehouse distribution. (3) Bundle slow-moving inventory items with high-demand products in the Books and Clothing categories.

5. Step-by-Step Microsoft Excel Dashboard Implementation Blueprint

To recreate this dynamic analysis inside Microsoft Excel, structure your workbook according to the following guidelines:

- **Sheet 1: Raw_Data** - Contains the unedited CSV/Excel export.
- **Sheet 2: Cleaned_Data** - The operational table. Convert using Ctrl + T and name the table SalesData. Ensure OrderDate is explicitly formatted as a Date, and TotalAmount is formatted as Currency.
- **Sheet 3: Pivot_Tables** - Construct independent Pivot Tables referencing SalesData:
 - *Pivot 1:* Rows = State, Values = Sum of TotalAmount (Sorted Descending).
 - *Pivot 2:* Rows = Category, Values = Sum of TotalAmount.
 - *Pivot 3:* Rows = OrderDate (Grouped by Month), Values = Sum of TotalAmount.
 - *Pivot 4:* Rows = ProductName, Values = Sum of TotalAmount (Filtered via Top 10 Value Filter).
- **Sheet 4: Dashboard** - The visual reporting canvas. Set up KPI summary cards at the top using formulas (e.g., =SUM(SalesData[TotalAmount])). Insert corresponding PivotCharts: a Horizontal Bar Chart for States, a Pie/Donut Chart for Categories, and a Smooth Line Chart for Monthly Trends. Insert Slicers for State, Category, and PaymentMethod, and right-click them to choose **Report Connections** to link them to all Pivot Tables.